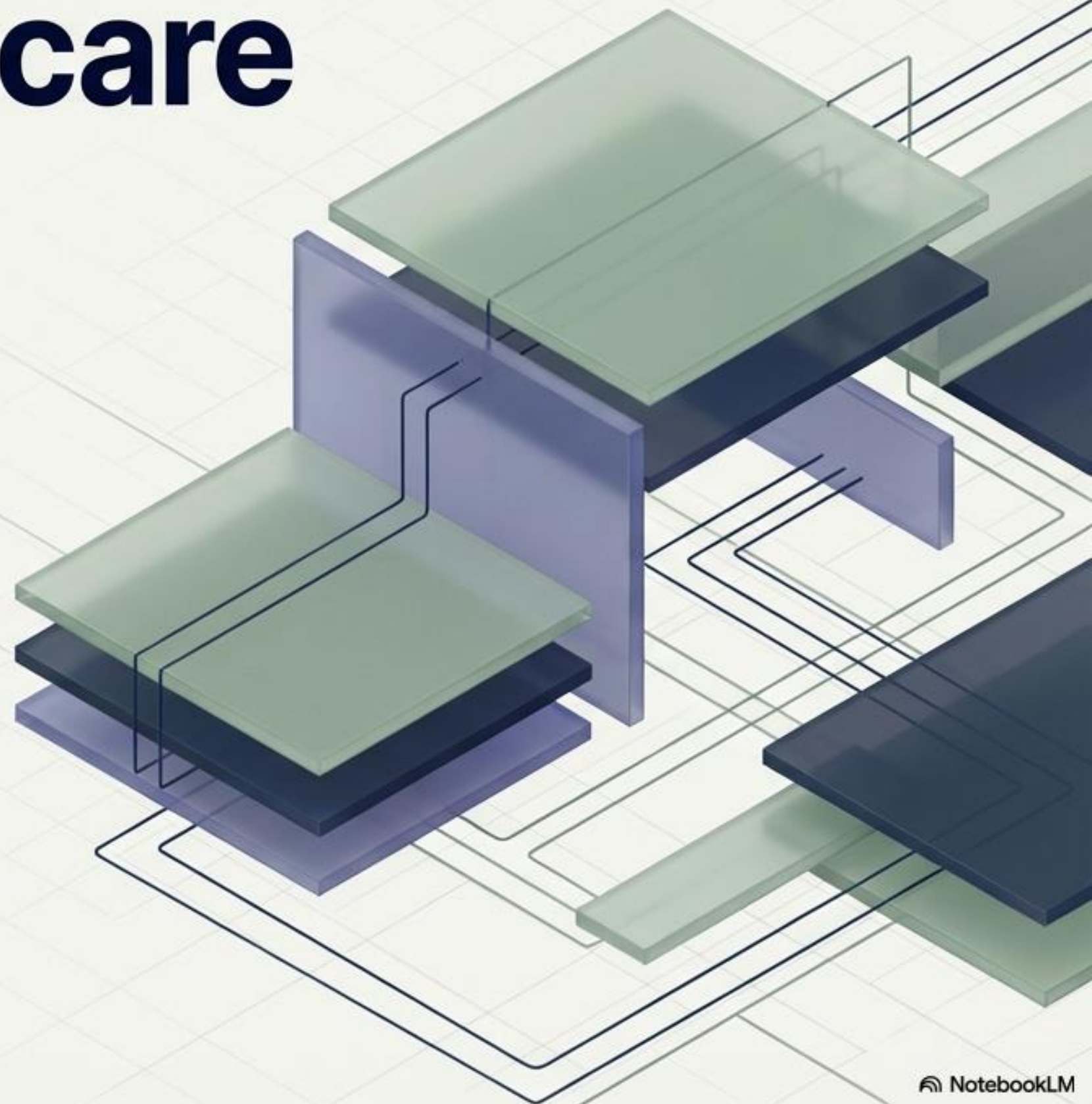


Building Healthcare Intelligence on Cortex Code

Architecture, Governance
Lifecycle, and Deployment

Health Sciences Industry Solutions Tiger Team | March 2026



The Layered Architecture of Healthcare Intelligence

Orchestrator Agent

Intent Detection → Domain Routing → Skill Composition

health-sciences-incubator.md

Enforces HIPAA guardrails across workflows

Domain Skills

17 Modular Health Sciences Skills

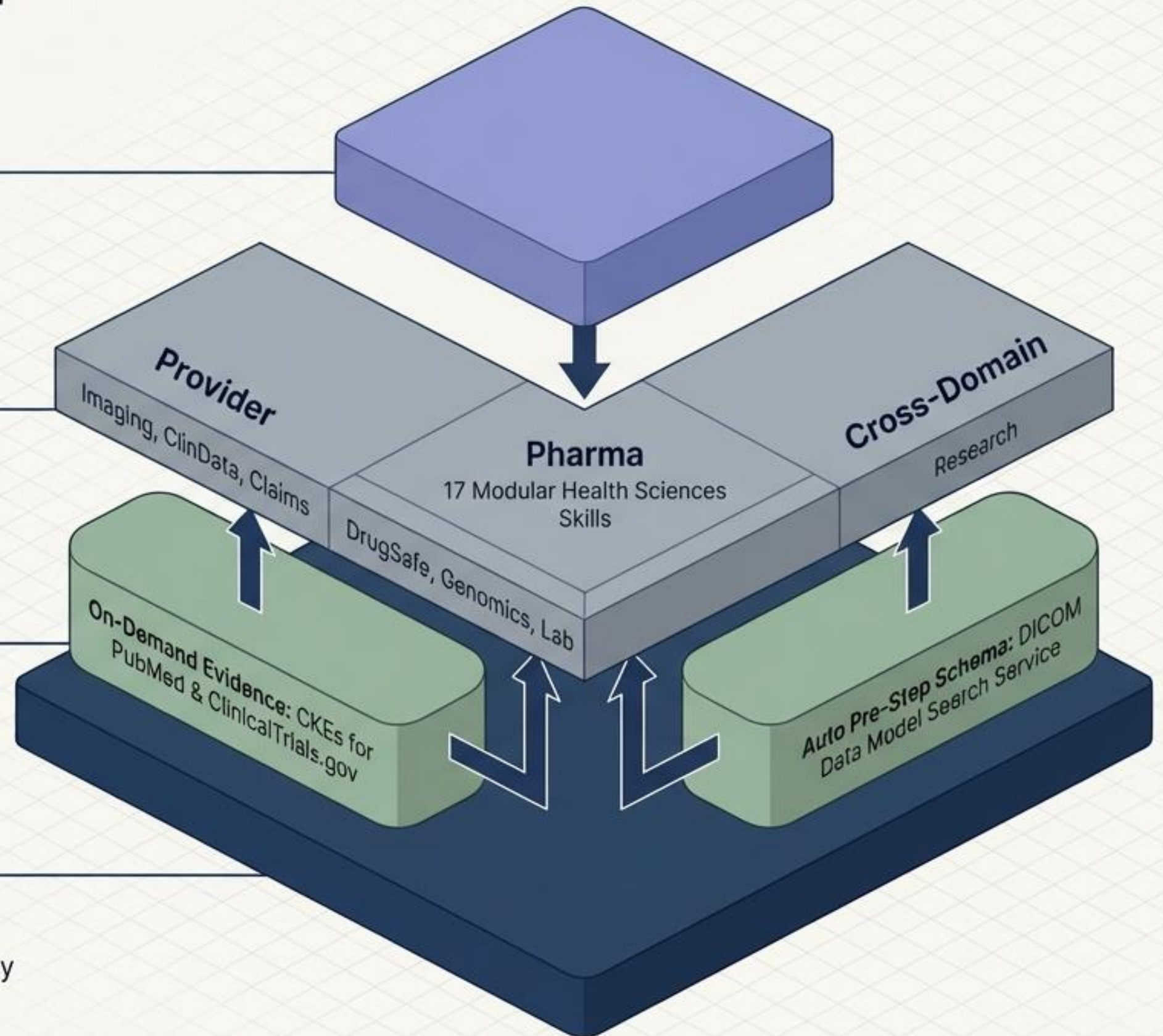
Knowledge Context

On-Demand Evidence: CKEs for PubMed
& ClinicalTrials.gov

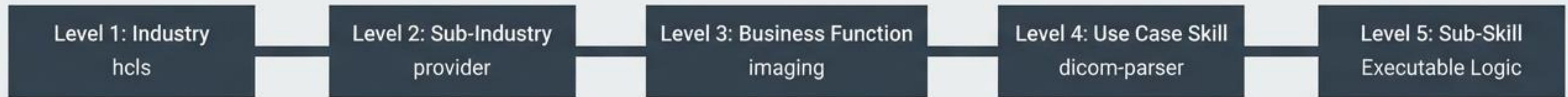
Snowflake Platform Foundation

25+ Bundled Native Tools

Dynamic Tables | Cortex AI | Streamlit | SPCS | dbt | ML Registry



Framework Structural Taxonomy



`hcls-provider-imaging-dicom-parser`

The hierarchy is encoded directly in the flat skill name, requiring exactly 1 level of nesting for auto-discovery.

The Healthcare Skills Collection

Provider
Domain

imaging

dicom-parser

fhir

clinical-nlp

omop

claims-data-
analysis

Pharma
Domain

pharmacovigilance

clinical-trial-
protocol

nextflow

variant-annotation

single-cell-qc

scvi-tools

survival-analysis

allotrope

Cross-Industry
& CKE

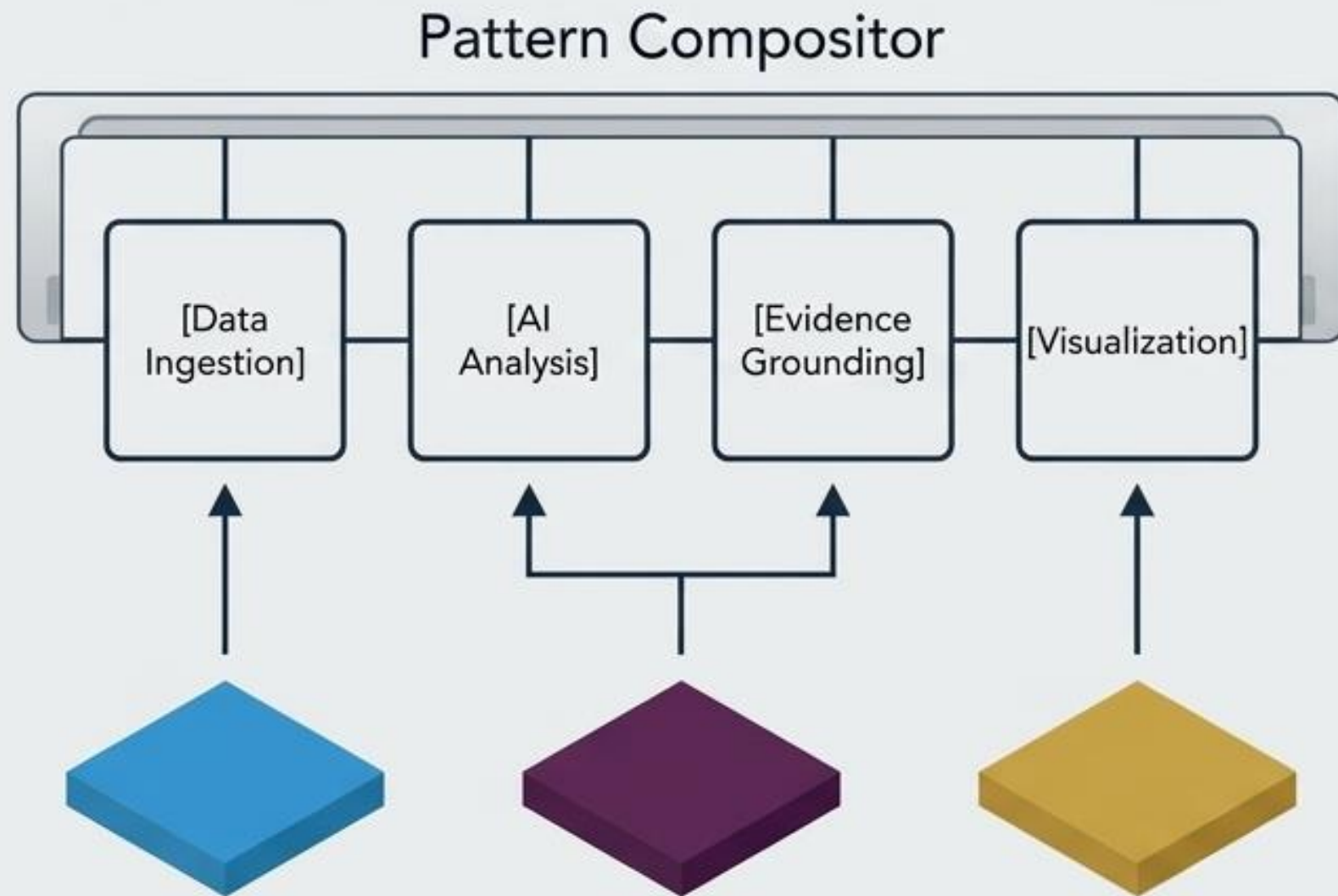
problem-selection

cke-pubmed

cke-clinical-trials

Framework
Inventory: 15
specialized
domain skills
+ 2 shared
Cortex
Knowledge
Extensions
(CKEs).
Automatically
invoked via
on-demand
routing.

Cross-Domain Solution Patterns



1. Imaging + Clinical Integration
2. Clinical Data Warehouse (OMOP)
3. Drug Safety Signal Detection
4. Genomics + Clinical Outcomes
5. Single-Cell Analysis Pipeline
6. Real-World Evidence Study
7. Clinical Trial Design
8. Lab Data Modernization
9. Clinical Data Application (React)

Natural Language Translation into Complex Taxonomy

Medical Imaging

I have DICOM files from our radiology department on S3. Build a complete pipeline to parse, ingest, and analyze the imaging metadata on Snowflake.



`hcls-provider-imaging`

(Auto-triggers dicom-parser and dicom-ingestion sub-skills)

Drug Safety

Analyze FDA FAERS data for adverse events associated with aspirin. Detect safety signals and cross-reference with published literature.



`hcls-pharma-dsafety-pharmacovigilance`

(Invokes on-demand PubMed CKE)

Genomics

I have whole-genome sequencing FASTQ files. Run variant calling and annotate pathogenic variants with ClinVar and gnomAD frequencies.



`hcls-pharma-genomics-nextflow`

Cross-Domain Real-World Evidence

Build a real-world evidence study: cohort from claims, standardize to OMOP, run survival analysis, and validate against published literature.

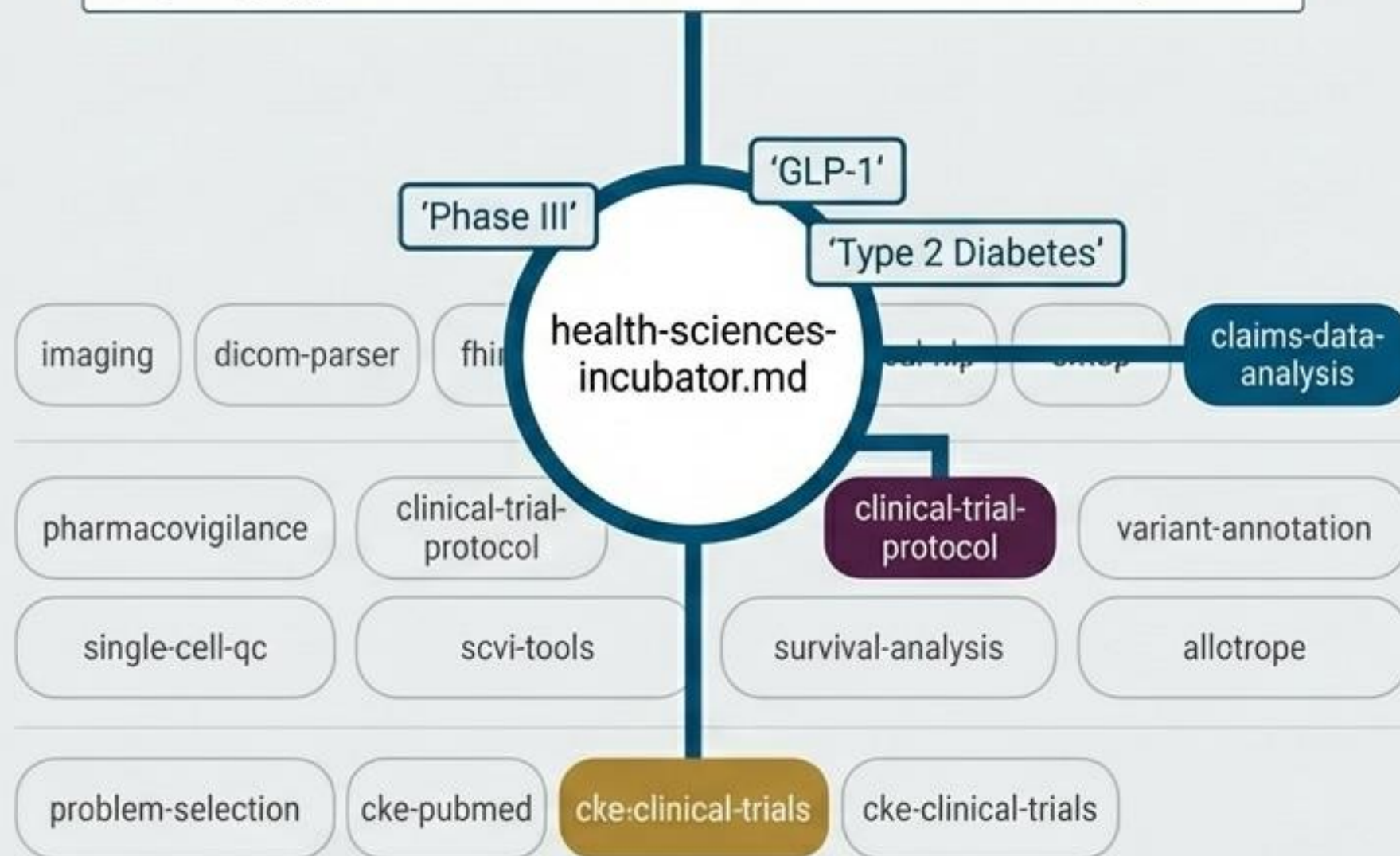


`Orchestrator Pattern Multi-Chain`

(Identifies OMOP CDM, Time-to-Event Outcomes, and Literature Grounding intents)

The Orchestrator's Canvas

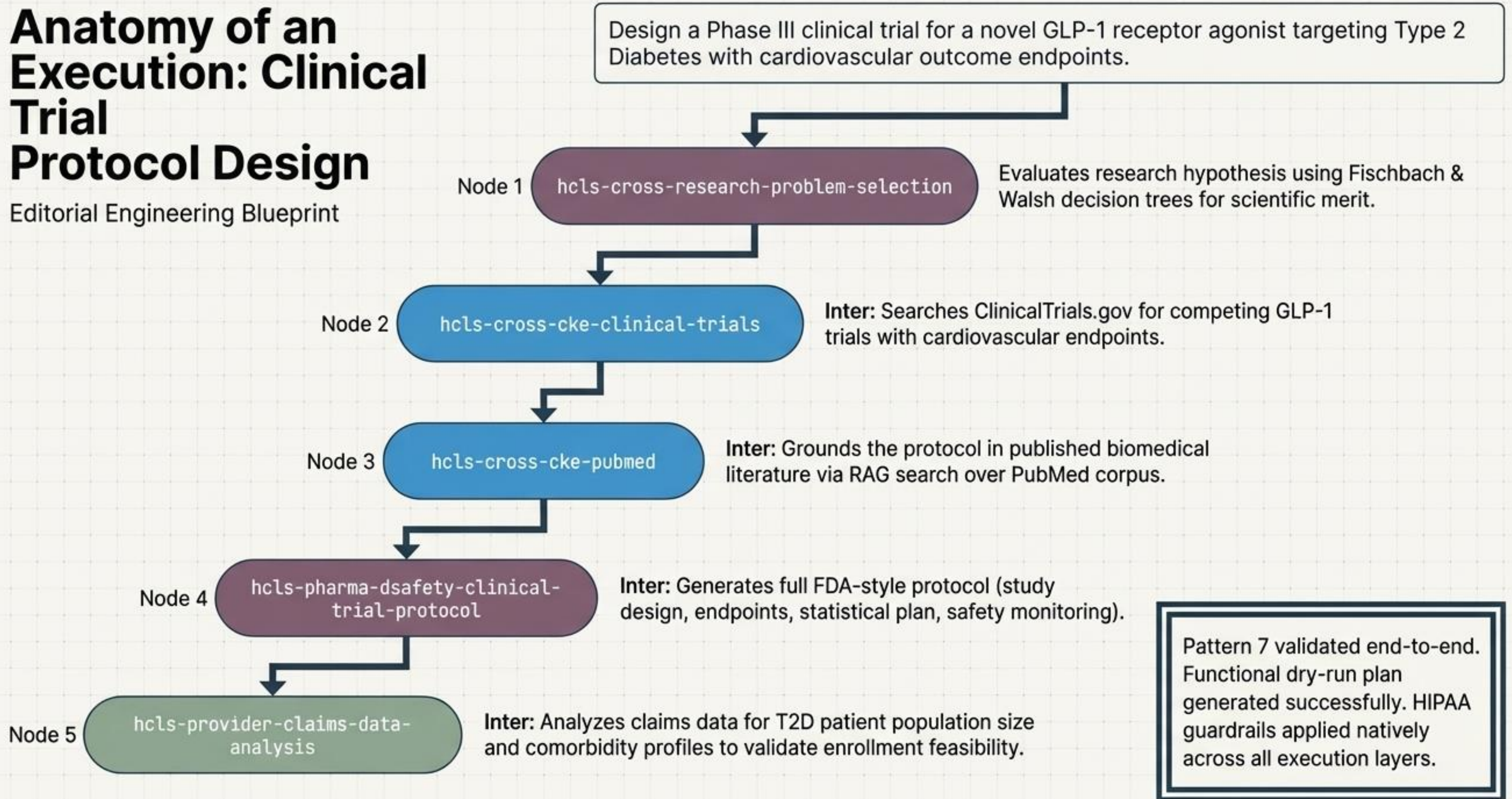
Design a Phase III clinical trial for a novel GLP-1 receptor agonist targeting Type 2 Diabetes with cardiovascular outcome endpoints.



Intent Detection → Domain Routing → Skill Composition.
The orchestrator automatically binds modular skills across boundaries to satisfy complex intent.

Anatomy of an Execution: Clinical Trial Protocol Design

Editorial Engineering Blueprint



Governed Velocity: The Two-Repo Development Lifecycle

Dual-Track Pipeline



Comparison Matrix

	Incubator Sandbox	Production SFS
GitHub Repository	Snowflake-Solutions/health-sciences-incubator	Snowflake-Solutions/cortex-code-skills
Contributor Access	Open (SEs, SAs, Field)	Restricted (Tiger Team Only)
Quality Bar	Low friction, rapid experimentation	High, formal PRs, staging gates
Cortex Profile	cortex profile add health-sciences-incubator	cortex profile add health-sciences-solutions

From Repository to Execution in Seconds

```
> git clone https://github.com/Snowflake-
Solutions/coco-healthcare-skills.git

> cortex profile add health-sciences-
incubator

> /skill
(Expected: 17 hcIs-* remote skills)

> /agents
(Expected: health-sciences-incubator loaded
via systemPromptRepo)
```

☐ Marketplace CKEs

Install PubMed & ClinicalTrials CKEs from Snowflake Marketplace. Unlocks on-demand RAG literature search for domain skills.

☐ Data Model Knowledge

Execute DICOM model script:
scripts/setup_dicom_model_knowledge_repo.sql
Unlocks live schema context for imaging tasks.

Keep Updated

Run **cortex skill update** to pull the latest GitHub changes.